

MH1301 Discrete Mathematics

Tutorial 1 — Problems & Solutions

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Overview of This Tutorial

Where pedagogically helpful, we:

- write counting arguments in explicit step-by-step form (product rule, sum rule, complement);
- use multiple viewpoints (strings $\leftrightarrow$ functions, direct counting $\leftrightarrow$ inclusion–exclusion);
- emphasize careful handling of edge cases (e.g. $n = 1$) and clear final answers.

Themes.

- **Ex. 6.1.3–6.1.4:** Basic counting via the product rule.
- **Ex. 6.1.6–6.1.7:** Bit strings; fixed endpoints; simple sums.
- **Ex. 6.1.9:** Counting strings with a required character; complement and inclusion–exclusion.
- **Ex. 6.1.10:** Counting RNA sequences under restrictions.
- **Ex. 6.1.12:** Counting integers in an interval; divisibility; inclusion–exclusion.
- **Ex. 6.1.13:** Digit strings; complement; exact counts.
- **Ex. 6.1.20–6.1.21:** Functions $A \rightarrow B$; injections; bijections to bit strings.

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Question 1 (Ex. 6.1.3 and Ex. 6.1.4)

Problem

- (a) Assume there are 26 letters. How many different three-letter initials can people have?
- (b) A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state. How many ways are there to form this committee?

Solution

Method 1: Product rule (multiplication principle)

- (a) For a three-letter initial, there are 3 positions to fill. For each position, there are 26 choices, and choices are independent across positions.

$$\# = 26 \cdot 26 \cdot 26 = 26^3.$$

Hence

$$\boxed{26^3 = 17576}.$$

- (b) For each of the 50 states, there are exactly 3 allowed choices for the representative (governor, senator 1, senator 2). Choices for different states are independent.

$$\# = \underbrace{3 \cdot 3 \cdots 3}_{50 \text{ times}} = 3^{50}.$$

Hence

$$\boxed{3^{50}}.$$

Method 2: Functions between finite sets

- (a) Let L be the set of 26 letters and let $P = \{1, 2, 3\}$ index the three positions. A three-letter initial is exactly a function $f : P \rightarrow L$ (choose a letter for each position). For each $i \in P$, there are $|L| = 26$ choices for $f(i)$, so

$$|\{f : P \rightarrow L\}| = 26^3,$$

and therefore

$$\boxed{26^3 = 17576}.$$

- (b) Let S be the set of 50 states and let $R = \{G, S_1, S_2\}$ be the 3 allowed roles (governor or the two senators). A committee selection is exactly a function $g : S \rightarrow R$ assigning to each state one of the 3 roles. Thus

$$|\{g : S \rightarrow R\}| = |R|^{|S|} = 3^{50},$$

so

$$\boxed{3^{50}}.$$

Question 2 (Ex. 6.1.6 and Ex. 6.1.7)

Problem

- (a) How many bit strings with length not exceeding n , where n is a positive integer, consist entirely of 1's, not counting the empty string?
- (b) How many bit strings of length n , where n is a positive integer, start and end with 1's?

Solution

Method 1: Direct enumeration / fixed-free positions

- (a) A bit string consisting entirely of 1's is uniquely determined by its length. For each length $\ell \in \{1, 2, \dots, n\}$, there is exactly one such string, namely $11 \cdots 1$ (with ℓ ones). Hence the count is

$$\# = n, \quad \boxed{n}.$$

- (b) If $n = 1$, then the only length-1 string that starts and ends with 1 is the string "1", so the count is 1.

Assume now $n \geq 2$. Then the first bit is fixed to be 1 and the last bit is fixed to be 1. The remaining $n - 2$ middle positions can each be chosen freely as 0 or 1, giving 2 choices per position. Thus

$$\# = 2^{n-2} \quad (n \geq 2).$$

Therefore,

$$\# = \begin{cases} 1, & n = 1, \\ 2^{n-2}, & n \geq 2. \end{cases}$$

Method 2: Sum rule and bijection to shorter strings

- (a) Let S_ℓ be the set of all-ones bit strings of length exactly ℓ . Then for each $\ell \geq 1$, $|S_\ell| = 1$. The desired set is the disjoint union

$$S_1 \dot{\cup} S_2 \dot{\cup} \cdots \dot{\cup} S_n.$$

By the sum rule,

$$\# = \sum_{\ell=1}^n |S_\ell| = \sum_{\ell=1}^n 1 = n, \quad \boxed{n}.$$

- (b) For $n \geq 2$, define a map φ from the set of length- n bit strings that start and end with 1 to the set of all bit strings of length $n - 2$ by deleting the first and last bits:

$$\varphi(1 b_2 b_3 \cdots b_{n-1} 1) := b_2 b_3 \cdots b_{n-1}.$$

This map is a bijection:

- (Injective) If two strings become equal after deleting the same fixed endpoints, then their middle blocks are equal, hence the original strings are equal.
- (Surjective) Given any length- $(n-2)$ string $c_1 \cdots c_{n-2}$, the length- n string $1c_1 \cdots c_{n-2}1$ maps to it.

So the number of such length- n strings equals the number of all length- $(n-2)$ bit strings, which is 2^{n-2} for $n \geq 2$. Together with the $n = 1$ case,

$$\# = \begin{cases} 1, & n = 1, \\ 2^{n-2}, & n \geq 2. \end{cases}$$

Question 3 (Ex. 6.1.9)

Problem

There are 128 different ASCII characters. How many strings of 5 ASCII characters contain the character '@' at least once?

Solution

Method 1: Complement counting

Let Ω be the set of all length-5 strings over the 128 ASCII characters. Then

$$|\Omega| = 128^5.$$

Let A be the set of length-5 strings that contain no '@'. For each of the 5 positions, there are 127 allowed characters (all but '@'), hence

$$|A| = 127^5.$$

The desired set is $\Omega \setminus A$, so

$$\# = |\Omega| - |A| = 128^5 - 127^5.$$

Compute the numerical value:

$$128^5 = (2^7)^5 = 2^{35} = 34359738368,$$

and

$$127^5 = 33038369407.$$

Therefore,

$$\boxed{128^5 - 127^5 = 1321368961}.$$

Method 2: Count by the number of '@' symbols

For $k \in \{1, 2, 3, 4, 5\}$, count strings with exactly k occurrences of '@'.

- Choose which k of the 5 positions are '@': $\binom{5}{k}$ ways.
- Fill the remaining $5 - k$ positions with any of the other 127 characters: 127^{5-k} ways.

Thus the total is

$$\# = \sum_{k=1}^5 \binom{5}{k} 127^{5-k}.$$

Using the binomial theorem,

$$\sum_{k=0}^5 \binom{5}{k} 127^{5-k} = (127 + 1)^5 = 128^5,$$

so

$$\sum_{k=1}^5 \binom{5}{k} 127^{5-k} = 128^5 - 127^5.$$

Hence

$$\boxed{128^5 - 127^5 = 1321368961}.$$

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Question 4 (Ex. 6.1.10)

Problem

There are 4 choices $\{G, U, A, C\}$ for each element of an RNA sequence. How many 6-element RNA sequences

- (a) do not contain U ?
- (b) end with GU ?
- (c) start with C ?
- (d) contain only A or U ?

Solution

Method 1: Product rule (independent positions)

- (a) If the sequence cannot contain U , then each of the 6 positions can be filled by one of $\{G, A, C\}$, i.e. 3 choices per position. Hence

$$3^6 = 729.$$

- (b) If the sequence must end with GU , then position 5 is fixed as G and position 6 is fixed as U . The first 4 positions are free, each with 4 choices. Hence

$$4^4 = 256.$$

- (c) If the sequence must start with C , then position 1 is fixed as C and the remaining 5 positions are free with 4 choices each. Hence

$$4^5 = 1024.$$

- (d) If the sequence can contain only A or U , then each position has 2 choices, so

$$2^6 = 64.$$

Method 2: Functions from positions to allowed alphabets

Let $P = \{1, 2, 3, 4, 5, 6\}$ be the position set. A length-6 RNA sequence is a function $f : P \rightarrow \{G, U, A, C\}$.

- (a) Restrict the codomain to $\{G, A, C\}$ (size 3). Then the number of functions is $3^{|P|} = 3^6$, hence

$$3^6 = 729.$$

- (b) Enforce $f(5) = G$ and $f(6) = U$. Then f is determined uniquely by its values on $\{1, 2, 3, 4\}$, and there are 4^4 possible restrictions to these four positions. Hence

$$4^4 = 256.$$

- (c) Enforce $f(1) = C$. Then f is determined by its restriction to $\{2, 3, 4, 5, 6\}$, which can be any function into a 4-element set, giving 4^5 . Hence

$$4^5 = 1024.$$

- (d) Restrict the codomain to $\{A, U\}$ (size 2). Then the number of functions is 2^6 , hence

$$2^6 = 64.$$

Question 5 (Ex. 6.1.12)

Problem

How many positive integers between 100 and 999 (both inclusive)

- (a) are divisible by 7?
- (b) are odd?
- (c) have the same three decimal digits?
- (d) are not divisible by 4?
- (e) are divisible by 3 or 4?
- (f) are not divisible by either 3 or 4?
- (g) are divisible by 3 and 4?
- (h) are divisible by 3 but not by 4?

Solution

Method 1: Arithmetic progressions and inclusion–exclusion

There are $999 - 100 + 1 = 900$ integers from 100 to 999 inclusive.

- (a) Multiples of 7 in $[100, 999]$:

$$7 \cdot 15 = 105 \geq 100 \quad (\text{first}), \quad 7 \cdot 142 = 994 \leq 999 \quad (\text{last}).$$

Thus there are

$$142 - 15 + 1 = 128$$

multiples, so

$$\boxed{128}.$$

- (b) The numbers 100–999 form 900 consecutive integers, starting with an even number (100). In any consecutive block of even length, exactly half are odd. Hence

$$\boxed{450}.$$

- (c) Numbers with the same three digits are

$$111, 222, \dots, 999,$$

which is 9 numbers, hence

$$\boxed{9}.$$

- (d) Count multiples of 4, then subtract from 900. The first multiple of 4 in the interval is $100 = 4 \cdot 25$, and the last is $996 = 4 \cdot 249$. Hence

$$\#\{\text{divisible by } 4\} = 249 - 25 + 1 = 225,$$

so

$$\#\{\text{not divisible by } 4\} = 900 - 225 = 675, \quad \boxed{675}.$$

- (e) Let A be divisible by 3, B be divisible by 4. First count $|A|, |B|, |A \cap B|$.

$$3 \cdot 34 = 102 \geq 100, \quad 3 \cdot 333 = 999 \leq 999 \Rightarrow |A| = 333 - 34 + 1 = 300.$$

Also $|B| = 225$ from part (d). Now $A \cap B$ is divisibility by $\text{lcm}(3, 4) = 12$:

$$12 \cdot 9 = 108 \geq 100, \quad 12 \cdot 83 = 996 \leq 999 \Rightarrow |A \cap B| = 83 - 9 + 1 = 75.$$

By inclusion-exclusion,

$$|A \cup B| = |A| + |B| - |A \cap B| = 300 + 225 - 75 = 450,$$

so

$$\boxed{450}.$$

- (f) “Not divisible by either 3 or 4” means complement of $A \cup B$ within the 900 numbers:

$$900 - |A \cup B| = 900 - 450 = 450, \quad \boxed{450}.$$

- (g) “Divisible by 3 and 4” means $A \cap B$, already computed as 75:

$$\boxed{75}.$$

- (h) “Divisible by 3 but not by 4” means $A \setminus B$, so

$$|A \setminus B| = |A| - |A \cap B| = 300 - 75 = 225, \quad \boxed{225}.$$

Method 2: Floor-function counting (systematic endpoint formulas)

For an integer $d \geq 1$, the number of multiples of d in $[100, 999]$ is

$$\left\lfloor \frac{999}{d} \right\rfloor - \left\lfloor \frac{99}{d} \right\rfloor,$$

because multiples of d up to 999 are $\lfloor 999/d \rfloor$, and multiples up to 99 (i.e. below 100) are $\lfloor 99/d \rfloor$.

- (a) For $d = 7$:

$$\left\lfloor \frac{999}{7} \right\rfloor - \left\lfloor \frac{99}{7} \right\rfloor = 142 - 14 = 128, \quad \boxed{128}.$$

(b) Odds in $[100, 999]$. The odd numbers form the arithmetic progression

$$101, 103, \dots, 999,$$

with first term 101, last term 999, common difference 2. The number of terms is

$$\frac{999 - 101}{2} + 1 = \frac{898}{2} + 1 = 449 + 1 = 450, \quad \boxed{450}.$$

(c) Same three digits are aaa for $a \in \{1, 2, \dots, 9\}$, giving 9 choices:

$$\boxed{9}.$$

(d) Not divisible by 4:

$$\#(\text{divisible by } 4) = \left\lfloor \frac{999}{4} \right\rfloor - \left\lfloor \frac{99}{4} \right\rfloor = 249 - 24 = 225,$$

so

$$\#(\text{not divisible by } 4) = 900 - 225 = 675, \quad \boxed{675}.$$

(e) Divisible by 3 or 4:

$$|A| = \left\lfloor \frac{999}{3} \right\rfloor - \left\lfloor \frac{99}{3} \right\rfloor = 333 - 33 = 300,$$

$$|B| = \left\lfloor \frac{999}{4} \right\rfloor - \left\lfloor \frac{99}{4} \right\rfloor = 249 - 24 = 225,$$

$$|A \cap B| = \left\lfloor \frac{999}{12} \right\rfloor - \left\lfloor \frac{99}{12} \right\rfloor = 83 - 8 = 75.$$

Thus

$$|A \cup B| = 300 + 225 - 75 = 450, \quad \boxed{450}.$$

(f) Not divisible by either 3 or 4:

$$900 - |A \cup B| = 900 - 450 = 450, \quad \boxed{450}.$$

(g) Divisible by 3 and 4:

$$|A \cap B| = 75, \quad \boxed{75}.$$

(h) Divisible by 3 but not by 4:

$$|A| - |A \cap B| = 300 - 75 = 225, \quad \boxed{225}.$$

Question 6 (Ex. 6.1.13)

Problem

How many strings of three decimal digits

- (a) do not contain the same digit three times?
- (b) begin with an odd digit?
- (c) have exactly two digits that are 4's?

Solution

Method 1: Direct counting (product rule, complement, and cases)

A string of three decimal digits is a length-3 string over $\{0, 1, \dots, 9\}$, so there are $10^3 = 1000$ total.

- (a) Exclude the 10 strings with all digits equal:

$$000, 111, \dots, 999.$$

Hence

$$\# = 1000 - 10 = 990, \quad \boxed{990}.$$

- (b) The first digit must be odd, so it can be one of $\{1, 3, 5, 7, 9\}$ (5 choices). The remaining two digits are arbitrary (10 choices each). Hence

$$\# = 5 \cdot 10 \cdot 10 = 500, \quad \boxed{500}.$$

- (c) Exactly two digits are 4's:

- Choose which two of the three positions are 4: $\binom{3}{2} = 3$.
- The remaining position must be a digit different from 4: 9 choices.

Thus

$$\# = 3 \cdot 9 = 27, \quad \boxed{27}.$$

Method 2: Equivalent formulations via sets of strings

Let $\Omega = \{0, 1, \dots, 9\}^3$ be the set of all length-3 digit strings, so $|\Omega| = 10^3 = 1000$.

- (a) Let $E = \{(d, d, d) : d \in \{0, 1, \dots, 9\}\}$ be the set of strings with all three digits equal. Then $|E| = 10$. The desired set is $\Omega \setminus E$, hence

$$\# = |\Omega| - |E| = 1000 - 10 = 990, \quad \boxed{990}.$$

- (b) Let $O = \{1, 3, 5, 7, 9\}$ be the set of odd digits. The set of strings beginning with an odd digit is $O \times \{0, \dots, 9\} \times \{0, \dots, 9\}$, whose cardinality is

$$|O| \cdot 10 \cdot 10 = 5 \cdot 10^2 = 500, \quad \boxed{500}.$$

- (c) Let S be the set of strings with exactly two 4's. For each choice of the unique position that is *not* 4 (3 choices), the non-4 digit can be any of the 9 digits in $\{0, \dots, 9\} \setminus \{4\}$. Therefore

$$|S| = 3 \cdot 9 = 27, \quad \boxed{27}.$$

Question 7 (Ex. 6.1.20 and Ex. 6.1.21)

Problem

Consider functions that map from the set $A = \{1, 2, \dots, n\}$ to the set $B = \{0, 1\}$, where n is a positive integer.

- How many such functions are there?
- How many such functions are one-to-one?
- How many such functions are there that assign 0 to both 1 and n ?
- How many such functions are there that assign 1 to exactly one element in A ?

Solution

Method 1: View functions as bit strings

A function $f : A \rightarrow B$ corresponds uniquely to a length- n bit string $(f(1), f(2), \dots, f(n))$.

- Each of the n positions can be 0 or 1, so there are 2^n bit strings, hence

$$\boxed{2^n}.$$

- A function $f : A \rightarrow \{0, 1\}$ is injective exactly when no two different elements of A map to the same value.
 - If $n = 1$, every function is injective, so there are 2.
 - If $n = 2$, injective means the two outputs differ, so the only possibilities are $(0, 1)$ and $(1, 0)$, i.e. 2.
 - If $n \geq 3$, injection is impossible because there are only 2 possible values but at least 3 inputs (pigeonhole principle).

Therefore,

$$\boxed{\# = \begin{cases} 2, & n = 1, \\ 2, & n = 2, \\ 0, & n \geq 3. \end{cases}}$$

- Require $f(1) = 0$ and $f(n) = 0$.
 - If $n = 1$, then $1 = n$ so the constraint is just $f(1) = 0$, giving 1 function.
 - If $n \geq 2$, then two positions are fixed as 0, and the remaining $n - 2$ positions are free bits, giving 2^{n-2} functions.

Hence

$$\# = \begin{cases} 1, & n = 1, \\ 2^{n-2}, & n \geq 2. \end{cases}$$

- (d) “Assign 1 to exactly one element” means the bit string has exactly one 1. Choose which index $i \in \{1, \dots, n\}$ has value 1 (there are n choices), and all others are forced to be 0. Hence

$$\boxed{n}.$$

Method 2: Direct function counting and injectivity constraints

- (a) For each input $i \in A$, the value $f(i)$ can be chosen independently as 0 or 1 (2 choices). By the product rule,

$$\# = \underbrace{2 \cdot 2 \cdots 2}_{n \text{ times}} = 2^n, \quad \boxed{2^n}.$$

- (b) An injective function $f : A \rightarrow B$ can exist only if $|A| \leq |B|$. Since $|A| = n$ and $|B| = 2$, injectivity forces $n \leq 2$ (pigeonhole principle).

- If $n = 1$, there are 2 functions total, and all are injective.
- If $n = 2$, injective functions are exactly bijections from a 2-element set to a 2-element set; there are $2! = 2$.
- If $n \geq 3$, there are no injective functions.

Thus

$$\# = \begin{cases} 2, & n = 1, \\ 2, & n = 2, \\ 0, & n \geq 3. \end{cases}$$

- (c) Enforce $f(1) = 0$ and $f(n) = 0$.

- If $n = 1$, the condition reduces to $f(1) = 0$, so there is 1 function.
- If $n \geq 2$, then for each of the remaining $n - 2$ inputs, there are 2 independent choices.

Hence

$$\# = \begin{cases} 1, & n = 1, \\ 2^{n-2}, & n \geq 2. \end{cases}$$

- (d) Exactly one element maps to 1 means the preimage $f^{-1}(\{1\})$ has size 1. Choose the unique element $i \in A$ that maps to 1 (there are n choices); then all other elements must map to 0, yielding a unique function per choice. Therefore

$$\boxed{n}.$$